

1. Administrative & Study Identification

Full Study Title: A Two-Part, Open-Label Systemic Gene Delivery Study to Evaluate the Safety and Expression of R07494222 (SRP-9001) in Subjects Under the Age of Four With Duchenne Muscular Dystrophy **Short Title/Acronym:** ENVOL **Protocol Number:** BN43881 / Study 302 **NCT Number:** NCT06128564 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Delandistrogene moxeparvovec (R07494222 / SRP-9001 / Elevidys™) **Phase of Development:** Phase II **Medical Monitor:** Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety of the systemically delivered gene therapy, delandistrogene moxeparvovec, in participants with Duchenne Muscular Dystrophy (DMD) under 4 years of age. **Secondary Objectives:** To evaluate the expression of delandistrogene moxeparvovec micro-dystrophin protein following the gene transfer.

2.2 Background & Rationale To address the progressive loss of muscle function in Duchenne Muscular Dystrophy caused by the absence of functional dystrophin protein. DMD is a rare, fatal neuromuscular disorder. Administering the single-dose AAV-based gene therapy to children under the age of four (including newborns and babies) aims to intervene early in the disease process, delivering a micro-dystrophin gene to preserve muscle fibers, reduce progressive scarring, and alter the long-term clinical trajectory of the disease.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **SN\$:** 21 participants **Number of Sites:** Global multi-center trial (including sites in Germany, Italy, Spain, and the UK). **Estimated Study Start:** 29-Nov-2023 **Estimated Primary Completion:** November 2032 (approx. 264 weeks of follow-up per participant)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants under 4 years of age (Cohorts A: ≥ 3 to < 4 years, B: ≥ 2 to < 3 years, C:

\$> 6\$ months to \$< 2\$ years, D: \$\le 6\$ months). 2. Has a definitive diagnosis of DMD prior to screening based on documentation of clinical findings and prior confirmatory genetic testing using a clinical diagnostic genetic test. 3. A pathogenic frameshift mutation or premature stop codon contained between exons 18 and 79 (inclusive). 4. Able to cooperate with age-appropriate motor assessment testing.

4.2 Exclusion Criteria 1. Elevated Recombinant Adeno-Associated Virus Serotype rh74 (rAAVrh74) antibody titers as per protocol-specified criteria. 2. Exposure to gene therapy, investigational medication, or any treatment designed to increase dystrophin expression, within protocol-specified time limits. 3. Receiving regular oral corticosteroids as a treatment for DMD or planning to receive oral corticosteroids as a treatment for DMD within 1 year of baseline.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of 1.33×10^{14} vg/kg delandistrogene moxeparvovec. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** One-time, single-dose infusion, followed by a 264-week (5-year) observation period.

5.2 Concomitant & Prohibited Therapy Permitted: Protocol-mandated peri-infusion prophylactic corticosteroids to suppress potential immune responses to the AAV vector. **Prohibited:** Regular long-term oral corticosteroids for DMD within 1 year of baseline, or previous gene-therapy/dystrophin-altering treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -30 to 0 Informed Consent, DMD Genetic Confirmation, anti-AAVrh74 Antibody Screen
Baseline	Day 0	Single IV Infusion of Delandistrogene Moxeparvovec, Vitals, Safety Checks	Interim Week 12 to 260
Muscle Biopsy (for micro-dystrophin expression), Safety Labs (Hepatic, Cardiac), Motor Assessments	End of Study	Week 264	Final Safety Evaluations, End of 5-year Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants with a Treatment-emergent Adverse Event (TEAE), Serious Adverse Event (SAE), and Adverse Event of Special Interest (AESI). **Measurement Method:** Clinical observation, safety laboratory parameters, and continuous event recording over the study period.

7.2 Secondary & Exploratory Endpoints Definition: Change in Quantity of Delandistrogene Moxeparvovec Dystrophin. **Measurement Method:** Measured by Western Blot of biopsied muscle tissue (to confirm structural expression of the micro-dystrophin protein).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extensive monitoring for gene therapy-associated risks, including acute liver toxicity, immune-mediated myositis, myocarditis, and thrombocytopenia. **Serious Adverse Events (SAEs):** Routine 24-hour reporting protocols, monitored rigorously by regulatory authorities (e.g., EMA). **Data Monitoring Committee (DMC):** Independent safety tracking to pause or advance dosing based on safety signals observed post-infusion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive any amount of the study gene therapy). **Sample Size Justification:** \$N=21\$ is designed to provide sufficient biological expression and safety data for this ultra-rare pediatric subgroup without requiring a statistically powered efficacy control group. **Handling of Missing Data:** Standard handling for dropouts; expression data analyzed on an observed-case basis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring post-infusion, extending to remote tracking over the 5-year follow-up window. **Audit Trail:** Robust centralized lab evaluations for anti-AAVrh74 antibody screening using the Elecsys® assay and eCRF data locking protocols.

11. Study Identifiers & Governance

Primary ID: {{BN43881}} **Registry Number:** {{NCT06128564}} **Sponsor/Lead Organization:** {{Hoffmann-La Roche}} **Study Title:** {{ENVOL}} **Official Regulatory Title:** {{A Two-Part, Open-Label Systemic Gene Delivery Study to Evaluate the Safety and Expression of R07494222 (SRP-9001) in Subjects Under the Age of Four With Duchenne Muscular Dystrophy}}

12. Clinical Framework (DMD Specific)

Primary Condition: {{Duchenne Muscular Dystrophy (DMD)}}
Diagnosis Threshold: {{Definitive genetic confirmation of DMD mutation between exons 18 and 79}} **Medical Methodology:** {{AAVrh74-mediated Micro-Dystrophin Gene Therapy}} **Optimization Target:** {{Restoration of micro-dystrophin protein expression in muscle tissue}}

13. Study Procedures & Methodology

Management Pathway: {{Single-dose IV systemic gene delivery accompanied by prophylactic immunosuppression}} **Education Protocol (HCP):** {{Specialized site training on rAAVrh74 screening, gene therapy infusion, and hepatic/immune monitoring}} **Education Protocol (Patient):** {{Caregiver education on post-infusion isolation, symptom tracking, and strict medication compliance}} **Quality Audit Mechanism:** {{Standardized central lab processing for muscle biopsies and antibody quantification}}

14. Observation & Follow-Up Schedule

Total Duration: {{264 Follow-up weeks (5 years)}} **Onsite Visit Cadence:** {{Baseline, Week 12 (Biopsy), and regular protocol-specified follow-up milestones}} **Remote Monitoring Frequency:** {{Standard protocol remote safety check-ins}} **Checklist Review Interval:** {{Routine monitoring for liver enzymes, cardiac function, and motor capability}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				